
Automotive SoC Motor Driver

1.0 DATA ACQUISITION UNIT (DAU)

The data acquisition unit (DAU) captures voltages within and external to the device. The values of the captured voltages can then be read from **result** registers for further analysis. The DAU has 16 different channels which can be set up independently and connected to different voltage sources. This will minimise the CPU effort substantially as all the measurement from different sources are done by hardware and it is possible to read the measured data all at once. When a conversion is complete, DAU will generate a conversion complete flag or request interrupt service to NVIC when the DAU interrupt is enabled.

1.1 Features

- A 12-bit SAR (successive approximation) ADC with 16 selectable sources
- 1µs conversion speed
- Programmable conversion sequences
- Programmable conversion speed
- Programmable track and hold period
- Selectable trigger source
- Selectable trigger edge type
- GPIO voltage conversion
- Accessible Gate Drive Unit (GDU) analog information
- MCU die temperature acquisition
- Configurable interrupt triggers
- Internal and external conversion reference voltage
- APB Bus interface
- DMA interface support

1.2 Introduction

The DAU converts analog voltages external and internal to the system to digital signals which can be read and processed by the CPU. The DAU is a programable and configurable ADC unit. For example, the trigger source to initiate a conversion and measurement is programable and can be input from GPIO, PWMU, GTU, or forced from software. The DAU also has 16 independent source channels which can be set up to measure voltages on and off the device. The DAU also supports the DMA interface to transfer captured data to a desired memory location automatically to further minimize CPU burden.

1.3 Functional Description

The Functional block diagram of the DAU is shown in Figure 1-1. The DAU module has 16 slots, and these slots can be programmed independantly to measure a source voltage. There are in total sixteen different analog sources that can be assigned to each slot. The source is selected by programming the channel value in the `slotx_config` register for the corresponding slot. Some of the sources measured are internal to the device, such as the 3.3V regulator VD33. Other sources are external to the device, connected to the GPIO. The trigger source to sample the analog signals can be driven by software (SW_TRIG), GPIO , GTU (Tx_Trigger) or PWMU (PWM_Centre). When any of the GPIO pins, PD0 to PD7, are used as a trigger source, the corresponding pin needs to be configured as a digital input. When any of the GPIO pins, PD0 to PD7, are used as a measurement source, the corresponding pin needs to be configured as an analog input. (Please refer to the GPIO documentation to learn about configuring the pins). The trigger source edge detection is also selectable and can be rising edge, falling edge, or both. The reference voltage source to the ADC unit within the DAU can be either external V_{ext} , or internal V_{BG} . If the external source V_{ext} is desired, it should be connected to PD6. The reference ground source to the ADC unit within the DAU can be either external GND_{ext} , or internal GND_{INT} . The external ground GND_{ext} has to be connected to PD7. The DAU control logic will manipulate the incoming result from the ADC based on control and configuration settings, and store the information in the `DAU_RESULTx` registers. There are also status and diagnostic registers available to indicate a measurement completion or report potential issues with the measurement.

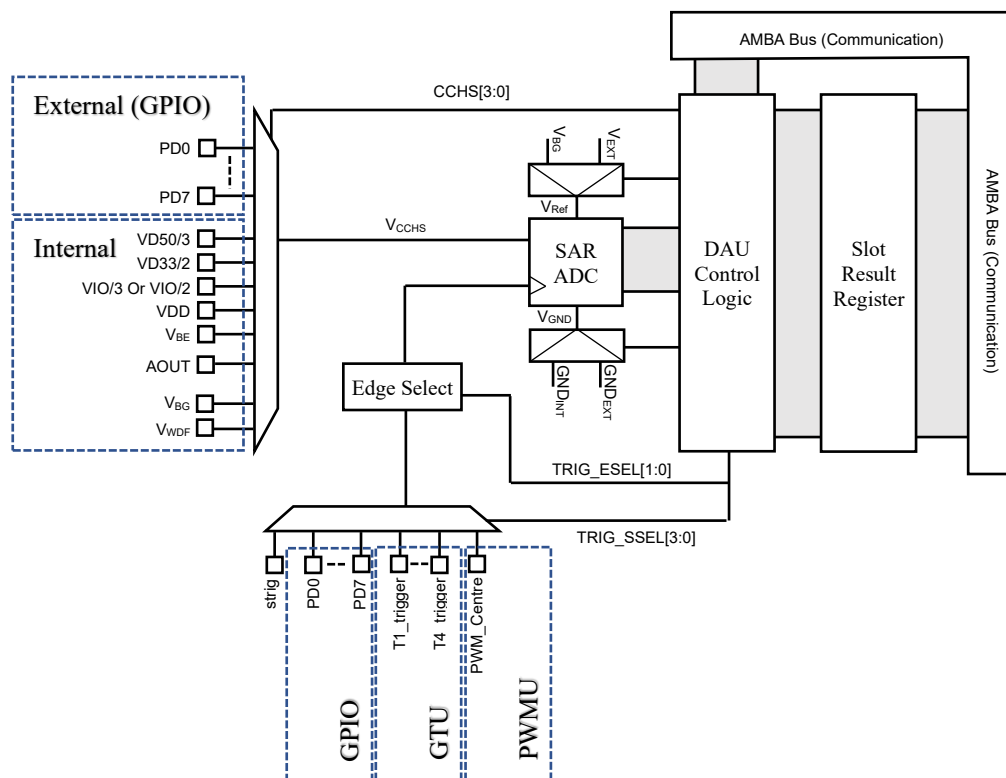


Figure 1-1: General block diagram of the DAU

1.4 Start-Up Sequence

It is possible to set-up the registers while the DAU is disabled, however, the module becomes functionally active once DAU_EN bit in DAU_CONTROL register is set to 1. After setting the DAU_EN bit to 1, the ADC shown in Figure 1-2 wakes up after DAU ADC wake up time, t_{aw} , and will be ready to implement the conversion. Once the ADC is powered up, ADCR in DAU_CONVERSION_STATUS register will be set to 1 to indicate the ADC is ready to implement the conversion. Figure 1-2 shows the start-up sequence for the DAU. The first trigger, at time A, is ignored because the DAU is not yet enabled (DAU_EN=0). The second trigger, at time B, is also ignored because although DAU_EN has been set to 1, the trigger at time B occurred sooner than t_{aw} , and the ADC has not yet finished powered up. TRIG_SKIP and ADC_ERR are flagged in the DAU_ERR_STATUS register to indicate the TRIG_SKIP has been due to ADC not fully turned on. At time C, the flags are reset by writing 1 to the corresponding flagged bits. The register settings need to be written before the valid trigger at time D occurs. Otherwise, the new settings will be effective only upon the next valid trigger signal. Once a valid trigger occurs, a minimum of TH_PRD+CNV_SPEED time is required before the result is ready to be read. When a conversion is completed for a slot, the corresponding SLOT_CC bit is set in the DAU_CONVCOMP_STATUS register, which can be read to determine if the new results are ready. It is also possible to generate interrupts if SLOT_ENCC is enabled in the DAU_INTSTATUS_CONVCOMP register. Finally the trigger at time E is ignored because a conversion is ongoing when the trigger occurs. In this case, TRIG_SKIP flag is set in DAU_ERR_STATUS register.

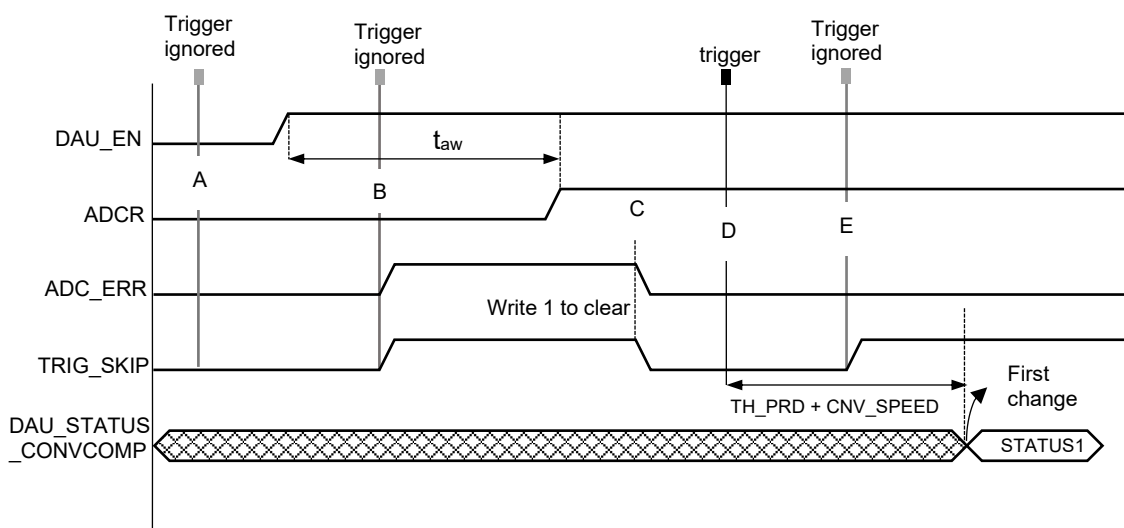


Figure 1-2: DAU start-up sequence

1.5 DAUMODs of operation

The DAU has 3 different modes of operation that can be selected by modifying the DAUMOD bits in the DAU_CONTROL register. This section explains each mode of operation. The following are the available modes of operation:

1. Single Source (DAUMOD = 0x0)
2. Common Trigger (DAUMOD = 0x1)
3. Sequential Trigger (DAUMOD = 0x2)

To activate the DAU, DAU_EN should be set to 1. When the DAU is enabled, 12bit SAR-ADC in Figure 1-1 will wake-up from sleep and will be ready to implement the conversion.

1.5.1 Single Source Mode

In the single source mode, the DAU carries out a conversion for only one channel slot and one input source per each trigger. This mode of operation can be achieved by setting DAUMOD in DAU_CONTROL register to 0. In this mode of operation, only slot0 is used and therefore the other 14 slot channels are not used. In this mode, only DAU_SLOT0_CONFIG register needs to be configured and the other config registers have no impact on the outcome. The result of the conversion is ultimately saved in DAU_RESULT0, once a valid trigger signal occurs after ADC power on.

The functional behavior of the DAU in single source mode is shown in Figure 1-3. At the start of a new trigger, DCNVS value in DAU_CONVERSION_STATUS register is changed to 10000_b to indicate slot0 is busy and the conversion is taking place for slot0. Slot0 remains busy for the duration defined by TH_PRD and CNV_SPEED. Once the conversion is finished DCNVS will change to 0 to indicate the DAU is ready to implement another conversion. SLOT_CC in DAU_CONVCOMP_STATUS register is set to 0001_H to indicate the conversion is complete for slot0. This bit can be cleared by writing a 1 to the corresponding bit in DAU_CONVCOMP_STATUS, which is shown at time E.

If any new trigger source occurs while the conversion is ongoing, the trigger will be ignored and the TRIG_SKIP bit will be flagged in the DAU_ERR_STATUS register. This behavior is demonstrated at time A and the flag can be cleared by writing a 1 to the TRIG_SKIP bit. If an attempt is made to change the configuration register while the conversion is on going, CONF_ERR will be flagged in DAU_ERR_STATUS register. This is shown at time B.

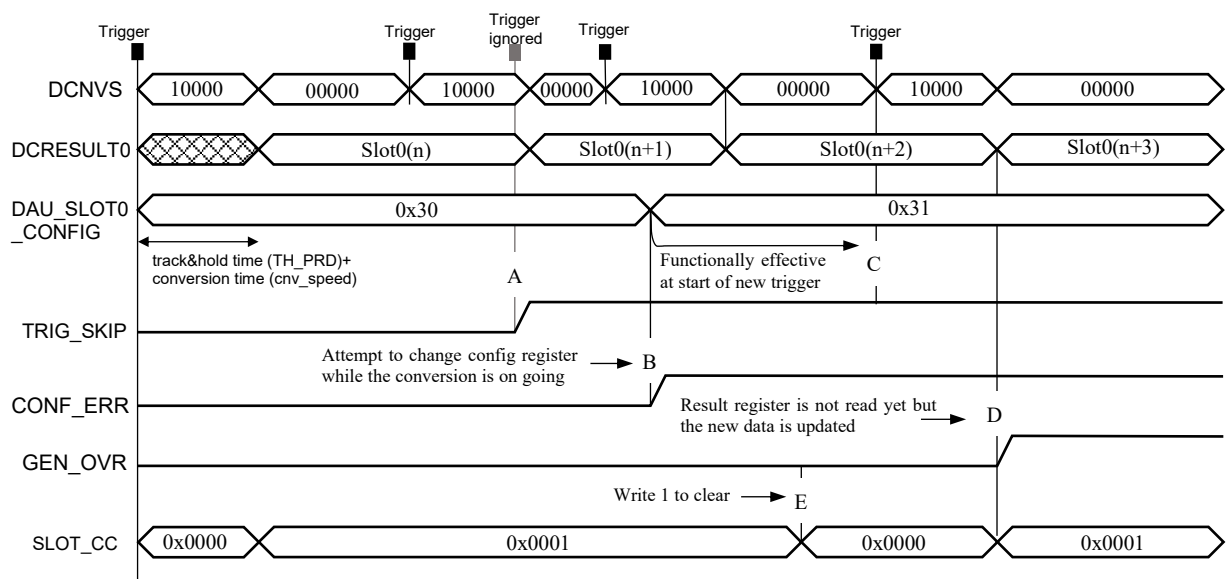


Figure 1-3: DAU Single Source Mode Functional Behaviour

When `DAU_SLOT0_CONFIG` or `DAU_CONTROL` registers are updated, the change in registers happens immediately, however, the new value is functionally effective at the start of the new trigger point which is marked as C in Figure 1-3. Conversion output, which is saved in the `DAU_RESULT0` register, must be read before a new value is written to `DAU_RESULT0`. If the value is not read and the `DAU_RESULT0` is overwritten with a new value, the `GEN_OVR` bit in the `DAU_ERR_STATUS` register and the `OVR_SLT` bit in the `DAU_OVR_STATUS` register will be set to 1. This behaviour is shown at time D in Figure 1-3. To clear `GEN_OVR`, the bits that are flagged in the `DAU_OVR_STATUS` register must first be cleared first by writing a 1 to them.

1.5.2 Sequential Trigger Mode

The DAU is set to sequential trigger mode by setting DAUMOD in DAU_CONTROL register to 010_b. Operation in sequential mode is similar to the single source mode except that the conversion slot automatically changes on every new trigger signal to the next valid conversion slot. This eliminates the need to re-program the settings in DAU_SLOTx_CONFIG registers on every new trigger signal, should different source measurement be required. The functional behavior of the DAU in sequential trigger mode is shown in Figure 1-4.

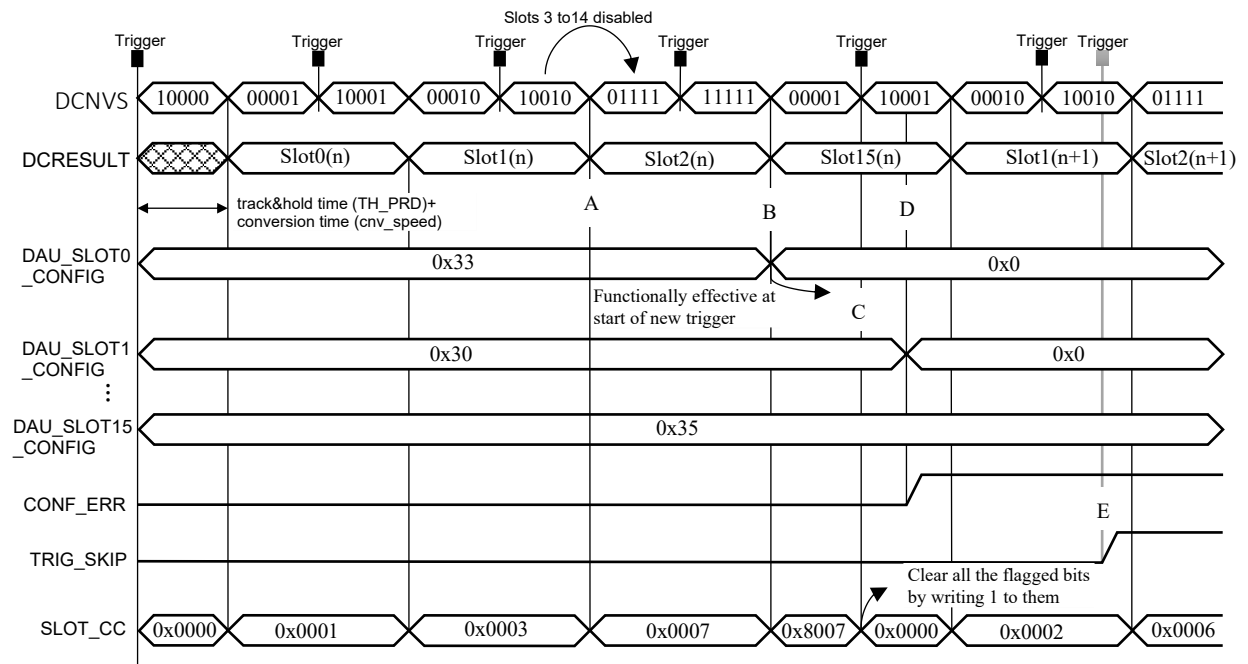


Figure 1-4: DAU Sequential Trigger Mode Functional Behavior

In sequential mode, the conversion starts from the first enabled slot and increments sequentially until it reaches the last enabled slot, and the sequence then repeats. If any of the slots are disabled between the first and last slot, they will be skipped. For example, in Figure 1-4, the conversion starts from the first enabled slot, which is slot0 in this case. It then sequences to slot1, slot2, and finally jumps to slot15, which is marked as time A. This is because the slots 3 through 14 are assumed to be disabled by setting SLOT_EN in the corresponding DAU_SLOTx_CONFIG registers to 0.

The DCNVS bits in the DAU_CONVERSION_STATUS register are constantly updated to indicate which channel is going to be converted next or which is currently being converted. For example, at time B, which is the end of first sequence, DCNVS indicates slot1 is scheduled for the next conversion instead of slot0. This is because at time B, SLOT_EN in DAU_SLOT0_CONFIG is set to '0' and slot0 is disabled. Therefore in the next trigger signal, which is marked as point C, the conversion starts with slot1.

The SLOT_CC in the DAU_CONVCOMP_STATUS register shows for which of the slots the conversion is completed. The bits in this register can be cleared by writing 1 to the flagged bits. For example, at time B, SLOT_CC value of 8007_H indicates that the conversion for the slots 0, 1, 2 and 15 are finished. At time E, these flagged bits are reset by writing 1 to them.

The DAU_SLOTx_CONFIG and DAU_CONTROL registers can be updated at any time, however, the change will be functionally effective only at the start of new trigger. For example, if the conversion channel source, CCHS, or track and hold time, TH_PRD, is changed, it is not going to take effect until the next trigger signal occurs. If any of the data_slot_x_config registers are updated while the conversion is ongoing, the change will be effective on the next trigger signal. However, the CONF_ERR will be flagged. The CONF_ERR can be cleared by writing 1 to CONF_ERR bit in DAU_ERR_STATUS register. For example at time D,

DAU_SLOT1_CONFIG is modified to disable slot1, however, the conversion for slot1 is ongoing and therefore CONF_ERR is flagged.

If a new trigger signal occurs while the sequence is not completed yet and the conversion is on going, the trigger is going to be skipped and TRIG_SKIP will be flagged in DAU_ERR_STATUS register which is shown at point E. The flag can be cleared by writing a 1 to TRIG_SKIP bit in the DAU_ERR_STATUS register.

Similar to single source trigger mode, if any of the DAU_RESULTx registers is not read before a new value is overwritten, the GEN_OVR bit is flagged. To clear the GEN_OVR flag, the bits that are flagged in DAU_OVR_STATUS register must first be cleared by writing a 1 to them.

1.5.3 Common Trigger Mode

The DAU is set to common trigger mode by setting DAUMOD in the DAU_CONTROL register to 001_b. In common trigger mode, the conversion for all the enabled slots is done upon each trigger signal. Therefore the difference between common trigger mode and sequential trigger mode is that in sequential mode only one slot is converted per each trigger signal where as in common trigger mode all the enabled slots are converted per each trigger signal.

The functional behaviour of the DAU in common trigger mode is shown in Figure 1-5. The conversion starts with the first enabled slot, which is slot0 in this case. The conversion sequence then continues until it reaches the last enabled slot which is slot 15, and is marked as time D in Figure 1-5. If any of the slots are disabled in between the first and last enabled slots, they will be skipped. For example, at time A, the trigger is skipped for slot1 because the slot is disabled by setting SLOT_EN in DAU_SLOT1_CONFIG to 0.

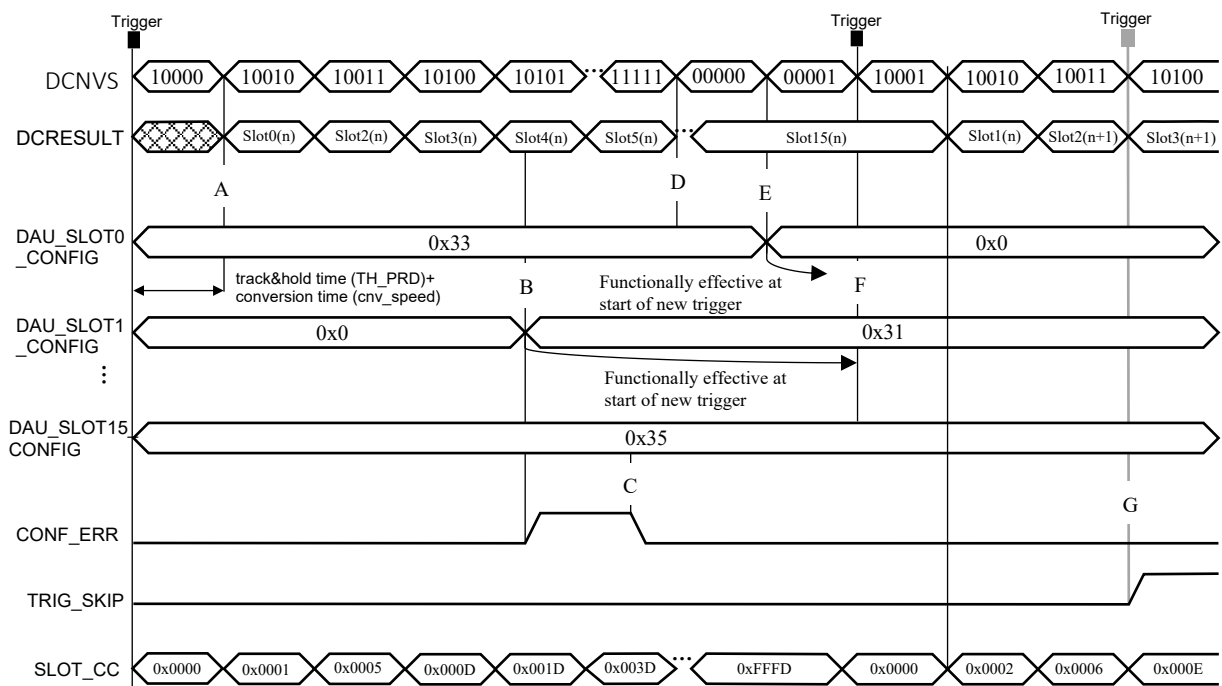


Figure 1-5: DAU Common Trigger DAUMOD Functional Behavior

The input registers can be updated at any time, however, they are functionally effective at the start of new trigger signal, which is marked as point F in Figure 1-5. If any of the DAU_SLOTx_CONFIG registers are changed while the conversion is ongoing, CONF_ERR will be flagged and the change will be functionally effective on the next trigger signal. For example, at time B, the DAU_SLOT1_CONFIG value is changed while the conversion is ongoing for slot5 and therefore CONF_ERR is flagged. The CONF_ERR flag can be reset by writing 1 to this bit in DAU_ERR_STATUS register. For example, at time C, a write of 1 is done to CONF_ER, and the flag is cleared.

The DCNVS bits in the DAU_CONVERSION_STATUS register are constantly updated to indicate for which slot a conversion is taking place or which is scheduled next. For example, at time D once the last conversion is complete, the DCNVS bits are changed to 0 to indicate slot0 is going to be sampled first once the next trigger occurs. However, at time E, slot0 is disabled and therefore DCNVS is changed to 00001_b to indicate the first slot going to be sampled with the next trigger is the slot1.

If a new trigger signal occurs while the sequence is not completed yet and the conversions are on going, the trigger is going to be skipped and TRIG_SKIP will be flagged in DAU_ERR_STATUS register which is shown at point G. The flag can be cleared by writing 1 to TRIG_SKIP bit in DAU_ERR_STATUS register.

Similar to single and sequential trigger modes, if DAU_RESULTx register is not read before a new value is updated, GEN_OVR bit will be flagged. To clear GEN_OVR flag, the bits that are flagged in DAU_OVR_STATUS register needs to be cleared by writing 1 to them.

1.6 Interrupt Management System

The DAU can generate interrupt requests to the CPU. The source of an interrupt request can be either due to an error or a conversion completion. There are enable signals for all the interrupt sources and therefore it is possible to target specific conditions to generate interrupts. Figure 1-6 shows the functional behaviour of the interrupt management system. All the interrupt request signals are passed to an OR gate in the final stage and an IRQ signal is generated to the NVIC system. The available interrupt sources are 16 conversion complete signals, SLOT_CC[x], 16 overrun signals, SLT_OVR[x], configuration error, CONF_ERR, trigger skip, TRIG_SKIP, and ADC error, ADC_ERR. All these interrupt sources have an associated enabled signal, and they can be disabled or enabled, depending on application requirements.

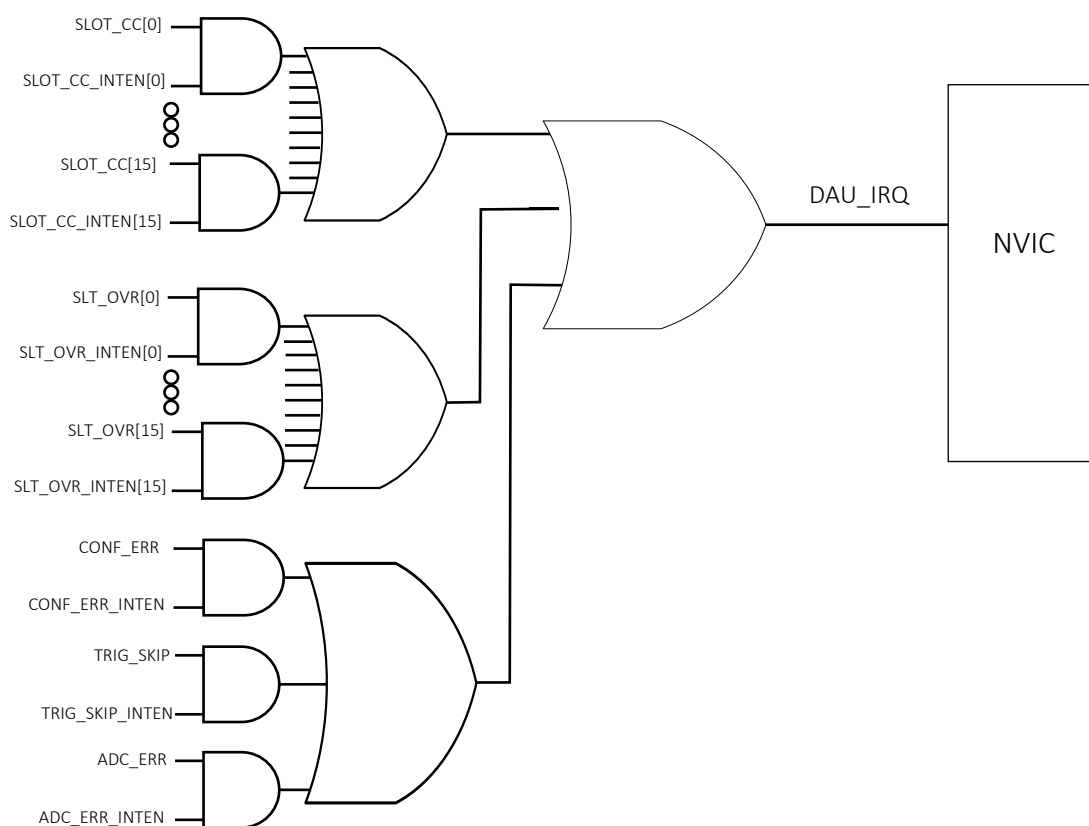


Figure 1-6: Interrupt Management System

1.7 Conversion Result

The output of the ADC in the DAU module is a 12-bit unsigned number, however, the result saved in the DAU_Resultx register is the ADC value shifted by two bits to the left. The least two significant and most significant bits are padded with 0. This is illustrated in Figure 1-7.

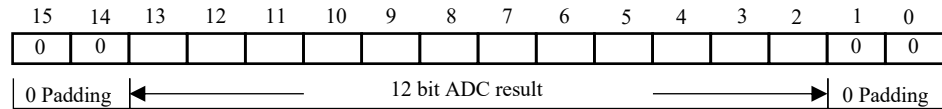


Figure 1-7: DAU_Resultx register after the shift

After reading the DAU_RESULTx register, shift it by 2 bit to the right to obtain DCRESULT[11:0].

The analog to digital conversion can be calculated as below:

$$DCRESULT[11:0] = 2048 \times \frac{V_{CCHS} - V_{GND}}{V_{Ref} - V_{GND}}$$

DCRESULT[11:0] is the DAU_RESULTx register value shifted by two bits to the right.

V_{GND} is the ground voltage, which can be either internal or external, depending on the DAU_GREFSEL setting in the DAU_CONTROL register. The internal V_{GND} is 0V.

V_{Ref} is the mid-rail voltage, which can be either internal or external, depending on the DAU_VREFSEL setting in the DAU_CONTROL register. The internal V_{Ref} is set to V_{BG} which is always 1.2V.

V_{CCHS} is the input voltage source to the ADC unit.

For example, if a voltage of 2V is applied to GPIO and is to be measured by the ADC unit, the calculation is performed as below (assuming internal V_{ref} and V_{GND}).

$$\begin{aligned} DCRESULT[11:0] &= 2048 \times \frac{V_{CCHS} - V_{GND}}{V_{Ref} - V_{GND}} \\ &= 2048 \times \frac{2 - 0}{1.2} = 3413 \end{aligned}$$

In this case, the register DAU_RESULTx value is set as follows:

$$DAU_RESULTx = DCRESULT[11:0] \times 4 = 3413 \times 4 = 13653$$

It should be noted that while the GPIO is 5V compliant, the maximum voltage that can be measured by input to GPIO is 2.4V. Any voltages higher than 2.4V, will be clamped to 2.4V for the purpose of measurement.

1.8 DMA Transfer

It is possible to transfer the results of the ADC conversion (DAU_RESULTx) by DAU to a specific memory location using DMA. This can minimise software overhead, as it is possible to record a long array of captured data before analysing it. When a conversion is completed, a DMA request is triggered. When the total number of conversions are equal to the number of DMA requests defined by MCCNUM in the DMA_DAU_REQNUM register, a DMA transfer will take place to transfer the data in DAU_RESULTx registers to a desired memory destination configured in the DMA. DMASLT_EN in the DMA_DAU_REQEN register defines which one of the slot conversions should contribute to the total number of DMA request. For example, if DMASLT_EN is set to 1110_b, then the conversion complete for slots 3, 2, and 1 will contribute to the total number of DMA requests. In this case, if MCCNUM is set to 3, then the DMA transfer will take place once a conversion is complete for slot 3. At the completion of the conversion for the last slot, slot15, the number of DMA requests will reset. Therefore, it is important that MCCNUM to be set to a value smaller or equal to DMASLT_EN, otherwise no DMA transfer will take place. For example, if DMASLT_EN is set to 0011_b, then the DMA request will be sent after the conversion is complete for slot0 and slot1, for a total of two DMA requests. In this case, if MCCNUM is set to a value greater than 2, a DMA transfer will not take place, since the total number of DMA requests will be reset at the completion of the conversion of last enabled slot, which in this case is slot2.

1.9 Register Set

The register map of the DAU consists of 16-bit registers. Table 1 summarizes the register addresses, access types and reset values. The starting address of the DAU is 0x40002000.

Base Address: 0x40002000

Table 1: Register map of the GTU

Register Name	Access Type	Address Offset	Reset Value
DAU CONTROL	R/W	0x00	0x0000
DAU SWTRIG	R/W	0x04	0x0000
DAU SLOT0 CONFIG	R/W	0x08	0x0000
DAU SLOT1 CONFIG	R/W	0x0C	0x0000
DAU SLOT2 CONFIG	R/W	0x10	0x0000
DAU SLOT3 CONFIG	R/W	0x14	0x0000
DAU SLOT4 CONFIG	R/W	0x18	0x0000
DAU SLOT5 CONFIG	R/W	0x1C	0x0000
DAU SLOT6 CONFIG	R/W	0x20	0x0000
DAU SLOT7 CONFIG	R/W	0x24	0x0000
DAU SLOT8 CONFIG	R/W	0x28	0x0000
DAU SLOT9 CONFIG	R/W	0x2C	0x0000
DAU SLOT10 CONFIG	R/W	0x30	0x0000
DAU SLOT11 CONFIG	R/W	0x34	0x0000
DAU SLOT12 CONFIG	R/W	0x38	0x0000
DAU SLOT13 CONFIG	R/W	0x3C	0x0000
DAU SLOT14 CONFIG	R/W	0x40	0x0000
DAU SLOT15 CONFIG	R/W	0x44	0x0000
DAU ERR STATUS	R/1C	0x48	0x0000
DAU ERR STATUS INTEN	R/W	0x4C	0x0000
DAU OVR STATUS	R/1C	0x50	0x0000
DAU OVR STATUS INTEN	R/W	0x54	0x0000
DAU CONVERSION STATUS	R	0x58	0x0000
DAU CONVCOMP STATUS	R/1C	0x5C	0x0000
DAU CONVCOMP STATUS INTEN	R/W	0x60	0x0000
DAU RESULT0	R	0x64	0x0000
DAU RESULT1	R	0x68	0x0000
DAU RESULT2	R	0x6C	0x0000
DAU RESULT3	R	0x70	0x0000
DAU RESULT4	R	0x74	0x0000
DAU RESULT5	R	0x78	0x0000
DAU RESULT6	R	0x7C	0x0000
DAU RESULT7	R	0x80	0x0000
DAU RESULT8	R	0x84	0x0000
DAU RESULT9	R	0x88	0x0000
DAU RESULT10	R	0x8C	0x0000
DAU RESULT11	R	0x90	0x0000
DAU RESULT12	R	0x94	0x0000
DAU RESULT13	R	0x98	0x0000
DAU RESULT14	R	0x9C	0x0000
DAU RESULT15	R	0xA0	0x0000
DAU DMA REQNUM	R/W	0xA4	0x0001
DAU DMA REQEN	R/W	0xA8	0x0000

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

DAU_CONTROL

Reset value = 0x0000

Register Address: 0x40002000

This register will define the control parameters for ADC operation mode, trigger edge selection, the source for triggering the measurement, ADC conversion speed, the mid-rail and GND reference voltage for ADC, and the master enable bit to power-up the ADC.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	DAU_EN	DAU_GREFSEL	DAU_VREFSEL	CNV_SPEED[1:0]		TRIG_SEL
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
	TRIG_SSEL[2:0]		TRIG_ESEL[1:0]		DAUMOD[2:0]		
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	
DAUMOD[2:0]	2:0	DAU Operation Mode. 000 _b : Single Source Mode 001 _b : Common Trigger Mode 010 _b : Sequential Trigger Mode The other settings are reserved.
TRIG_ESEL[1:0]	4:3	Trigger Edge Selection Selects the trigger edge that starts the conversion. 00: Rising 01: Falling 10: Rising or Falling 11: Rising
TRIG_SSEL[3:0]	8:5	Trigger Source Selection Select the input source that triggers the start of the conversion Input from Pulse Width Modulation UNIT (PWMU) 0000 _b : PWM_centre Software Trigger - the state of STRIG in the DAU_SWTRIG register is the trigger source. 0001 _b : SW_TRIG Input from General Timer Unit (GTU) 0010 _b : T1_Trigger 0011 _b : T2_Trigger 0100 _b : T2_Trigger 0101 _b : T3_Trigger Input from general purpose input port. The desired pin needs to be set as input in GPIO unit. 0110 _b : PD0 1010 _b : PD4 0111 _b : PD1 1011 _b : PD5 1000 _b : PD2 1100 _b : PD6 1001 _b : PD3 1101 _b : PD7 The other settings are reserved.

CNV_SPEED[1:0]	10:9	ADC Conversion Time 00: 1.0µs 01: 1.5µs 10: 2.0µs 11: 2.5µs
DAU_VREFSEL	11	DAU Reference Voltage Selection for the ADC Unit 0: Internal reference voltage 1: External reference voltage Note: The external reference voltage needs to be connected to PD6.
DAU_GREFSEL	12	DAU Reference Ground Selection for the ADC Unit 0: Internal Ground 1: External Ground Note: The external reference voltage needs to be connected to PD7.
DAU_EN	13	Master Enable Control Bit for DAU 0: Disabled 1: Enabled

DAU_SWTRIG

Reset value = 0x0000

Register Address: 0x40002004

When TRIG_SSEL is set to SW_TRIG in the DAU_CONTROL register, the STRIG bit in this register will act as the trigger source for ADC conversion.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	-	-	STRIG
U	U	U	U	U	U	U	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
STRIG	0	Software Trigger Control Bit This bit acts as the trigger source for ADC unit when TRIG_SSEL is set to SW_TRIG in DAU_CONTROL register.

DAU_SLOTx_CONFIG

Reset value = 0x0000

Register Address SLOT0: 0x40002008
 Register Address SLOT1: 0x4000200C
 Register Address SLOT2: 0x40002010
 Register Address SLOT3: 0x40002014
 Register Address SLOT4: 0x40002018
 Register Address SLOT5: 0x4000201C
 Register Address SLOT6: 0x40002020
 Register Address SLOT7: 0x40002024
 Register Address SLOT8: 0x40002028
 Register Address SLOT9: 0x4000202C
 Register Address SLOT10: 0x40002030
 Register Address SLOT11: 0x40002034
 Register Address SLOT12: 0x40002038
 Register Address SLOT13: 0x4000203C
 Register Address SLOT14: 0x40002040
 Register Address SLOT15: 0x40002044

There are 16 independant slots which can implement ADC conversions. These registers define the settings for each slot. The settings include the slot's conversion channel source, the track and hold period, and if the slot should be enabled or not to perform a conversion. When DAUMOD is set to '0' only slot0 is used and the configuration settings for slot1 through slot15 are ignored.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
TH_PRD[2:0]			SLOT_EN	CCHS[3:0]			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
CCHS[3:0]	3:0	ADC Conversion Channel Source Selects the source channel to be measured by the ADC unit. Input from general purpose input port. The desired pin needs to be set as input in the GPIO unit. 0000 _b : PD0 0010 _b : PD2 0100 _b : PD4 0110 _b : PD6 0001 _b : PD1 0011 _b : PD3 0101 _b : PD5 0111 _b : PD7 Internal 5V regulator (VD50), divided by 3. 1000 _b : VD50/3 Internal 3.3V regulator (VD33), divided by 2. 1001 _b : VD33/2 VIO voltage. This parameter is divided by 3 in 5V applications, and divided by 2 in 3.3V applications. 1010 _b : VIO/3 (5V application) : VIO/2 (3.3V application) Internal 1.2V regulator (VDD). 1011 _b : VDD

Voltage across diode indicating silicon temperature.
1100_b: V_{BE}

$V_{BE} = 0.7V - 2mV/DegC$
Where DegC is deviation from room temperature (23 °C)

Analog mux output (AOUT), which contains multiple signals from the GDU (gate drive unit). Refer to GDU section for further information
1101_b: AOUT

1.2V Bandgap voltage (V_{BG}), which is used as a reference signal for DAU.
1110_b: V_{BG}

Watchdog Clock Filtered:
1111_b: V_{WDF}

When the watchdog clock is working as expected, V_{WDF} varies between 0.4V to 0.8V

SLOT_EN	4	Enables the conversion for the slot 0: Disabled 1: Enabled	
TH_PRD[2:0]	7:5	track and hold period selection for the slot 000 _b : 1 μs 010 _b : 3 μs 100 _b : 10 μs 110 _b : 125 μs	001 _b : 2 μs 011 _b : 5 μs 101 _b : 25 μs 111 _b : 625 μs

DAU_ERR_STATUS

Reset value = 0x0000

Register Address : 0x40002048

This register will flag the potential errors that can occur during ADC conversion.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
GEN_OVR	-	-	-	-	-	-	-
R	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	CONF_ERR	ADC_ERR	TRIG_SKIP
R	U	U	U	U	R/1C	R/1C	R/1C

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
TRIG_SKIP	0	Trigger Skip Flag This bit is set when a trigger is missed due to an ongoing conversion, and when a trigger is missed due to the ADC being powered off. The ADC_ERR bit can be used to determine which error type caused this flag. 0: trigger was not missed 1: trigger was missed
ADC_ERR	1	ADC Error This bit is set when a trigger is missed due to the ADC being powered off. Thus, when there is TRIG_SKIP error, this bit can distinguish whether the error is due to an ongoing conversion or due to the ADC being powered off. 0: TRIG_SKIP is due to ongoing conversion 1: TRIG_SKIP is due to ADC power off
CONF_ERR	2	Configuration Error This bit is set when a conversion is ongoing and an attempt is made to change any of the input registers 0: No configuration error 1: Configuration error
GEN_OVR	15	General fault flag due to over-run When any of the over-run bits are flagged in DAU_OVR_STATUS register, the GEN_OVR bit will be flagged. The source of the GEN_OVR flag can be determined by reading DAU_OVR_STATUS register. Note: the GEN_OVR bit can only be cleared by clearing the corresponding bit in DAU_OVR_STATUS register.

DAU_ERR_STATUS_INTEN

Reset value = 0x0000

Register Address : 0x4000204C

This register will enable interrupt generation for errors that occur during ADC conversion.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	CONF_ERR_INTEN	ADC_ERR_INTEN	TRIG_SKIP_INTEN
U	U	U	U	U	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
TRIG_SKIP_INTEN	0	Trigger Skip Interrupt Enable Enables or disables the interrupt generation for TRIG_SKIP error. 0: Disabled 1: Enabled
ADC_ERR_INTEN	1	ADC Error Interrupt Enable Enables or disables the interrupt generation for ADC_ERR error. 0: Disabled 1: Enabled
CONF_ERR_INTEN	2	Configuration Error Interrupt Enable Enables or disables the interrupt generation for CONF_ERR error. 0: Disabled 1: Enabled

DAU_OVR_STATUS

Reset value = 0x0000

Register Address : 0x40002050

If the ADC results are ready in DAU_RESULTx registers and are not read before a new measurement is available, the over-run bit in this register corresponding to a DAU_RESULTx that has not been read will be set. For example, if DAU_RESULT1 is not read by the time a new measurement is available, OVR_SLT[15:0] is set to 0x0002.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
OVR_SLT[15:8]							
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
OVR_SLT[7:0]							
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
OVR_SLT[15:0]	15:0	Slot Over-run When a slot result register is overwritten before the previous data has been read, this flag will be asserted. 0x0000: There is no over-run in any of the slots. 0x0001: There is over-run in slot 0. 0x0002: There is over-run in slot 1. 0x0003: There is over-run in slot 1 and 0. ... 0xFFFF: There is over-run in all the slots.

DAU_OVR_STATUS_INTEN

Reset value = 0x0000

Register Address : 0x40002054

This register will enable the interrupt generation for the corresponding OVR_SLT[15:0] bits.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
OVR_SLT_INTEN[15:8]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
OVR_SLT_INTEN[7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
OVR_SLT_INTEN[15:0]	15:0	Slot Over-Run Interrupt Enable Enables the interrupt generation for the corresponding over-run condition. 0x0000: over-run Interrupt generation is disabled for all the slots. 0x0001: over-run Interrupt generation is enabled for slot 0. 0x0002: over-run Interrupt generation is enabled for slot 1. 0x0003: over-run Interrupt generation is enabled for slots 1 and 0. ... 0xFFFF: over-run Interrupt generation is enabled for all the slots.

DAU_CONVERSION_STATUS

Reset value = 0x0000

Register Address : 0x40002058

This register will indicate if a slot is busy, what slot is going to implement the next conversion, and also if the ADC is fully powered-up.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	ADCR	DCNVS[4:0]				
U	U	U	R	R	R	R	R

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
DCNVS[4:0]	4:0	DAU Conversion State. These bits indicate for which slot a conversion is ongoing or which is scheduled next. 00000: Ready, slot 0 conversion is scheduled next. 00001: Ready, slot 1 conversion is scheduled next. 00010: Ready, slot 2 conversion is scheduled next. ... 01111: Ready, slot 15 conversion is scheduled next. 10000: Busy, slot 0 conversion is ongoing. 10001: Busy, slot 1 conversion is ongoing. 10010: Busy, slot 2 conversion is ongoing. ... 11111: Busy, slot 15 conversion is ongoing.
ADCR	5	ADC Ready One bit to indicate if ADC is fully powered up. 0: ADC is disabled or powering-up 1: ADC is powered up

DAU_CONVCOMP_STATUS

Reset value = 0x0000

Register Address : 0x4000205C

This register will flag a corresponding bit once a conversion is completed for a slot channel. For example, if the conversion is complete for the slot0 and slot1, SLOT_CC[15:0] will be set to 0x0003.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
SLOT_CC[15:8]							
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
SLOT_CC[7:0]							
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
SLOT_CC[15:0]	15:0	Slot Conversion Complete These bits indicate if a conversion is complete for a corresponding slot. 0x0000: conversion is not complete for any of the slots. 0x0001: conversion is complete for slot 0. 0x0002: conversion is complete for slot 1. 0x0003: conversion is complete for slot 1 and 0. ... 0xFFFF: conversion is complete for all the slots.

Note: The bits in this register can be cleared by writing 1 to them.

DAU_CONVCOMP_STATUS_INTEN

Reset value = 0x0000

Register Address : 0x40002060

This register will enable the interrupt generation for a completed conversion (i.e. when the corresponding bit in DAU_CONVCOMP_STATUS register is set).

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
SLOT_CC_INTEN[15:8]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
SLOT_CC_INTEN[7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
SLOT_CC_INTEN[15:0]	15:0	Slot Conversion Complete Interrupt Enable Enables interrupt generation for conversion complete flag. 0x0000: conversion complete interrupt generation is disabled for all the slots. 0x0001: conversion complete interrupt generation is enabled for slot 0. 0x0002: conversion complete interrupt generation is enabled for slot 1. 0x0003: conversion complete interrupt generation is enabled for slot 1 and 0. ... 0xFFFF: conversion complete interrupt generation is enabled for all the slots.

DAU_RESULTx

Reset value = 0x0000

Register Address RESULT0: 0x40002064
 Register Address RESULT1: 0x40002068
 Register Address RESULT2: 0x4000206C
 Register Address RESULT3: 0x40002070
 Register Address RESULT4: 0x40002074
 Register Address RESULT5: 0x40002078
 Register Address RESULT6: 0x4000207C
 Register Address RESULT7: 0x40002080
 Register Address RESULT8: 0x40002084
 Register Address RESULT9: 0x40002088
 Register Address RESULT10: 0x4000208C
 Register Address RESULT11: 0x40002090
 Register Address RESULT12: 0x40002094
 Register Address RESULT13: 0x40002098
 Register Address RESULT14: 0x4000209C
 Register Address RESULT15: 0x400020A0

These registers will hold the ADC conversion output for the corresponding slot. For example, DAU_RESULT0 will record the conversion output for slot0, DAU_RESULT1 will record the conversion output for slot1, and so on.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
0	0	DCRESULT[11:6]					
R	R	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
DCRESULT[5:0]						0	0
R	R	R	R	R	R	R	R

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
DCRESULT[11:0]	15:0	DAU Conversion result Conversion result for the programmed slot.

$$DCRESULT[11:0] = 2048 \times \frac{V_{CCHS} - V_{GND}}{V_{Ref} - V_{GND}}$$

DCRESULT[11:0] is DAU_RESULTx register value shifted by two bit to the right. V_{GND} is the ground voltage which can be either internal or external, depending on the DAU_GREFSEL setting in DAU_CONTROL register. The internal V_{GND} is 0V. V_{Ref} is the mid-rail voltage which can be either internal or external, depending on the DAU_VREFSEL setting in DAU_CONTROL register. The internal V_{Ref} is set to V_{BG} which is always 1.2V. V_{CCHS} is the input voltage source to the ADC unit.

For example, if a voltage of 2V applied to GPIO to be measured by ADC unit the calculation can be done as below assuming internal V_{ref} and V_{GND} .

$$\begin{aligned}
 DCRESULT[11:0] &= 2048 \times \frac{V_{CCHS} - V_{GND}}{V_{Ref} - V_{GND}} \\
 &= 2048 \times \frac{2 - 0}{1.2} = 3413
 \end{aligned}$$

DAU_DMA_REQNUM

Reset value = 0x0001

Register Address: 0x400020A4

This register will define the minimum number of ADC conversions required to initiate a DMA transfer.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Res	Res	Res	MCCNUM[4:0]				
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
MCCNUM[4:0]	4:0	Minimum Conversion Complete Numbers for DMA Transfer Minimum number of conversions to be completed in order to initiate a DMA request to transfer conversion results from DAU_RESULTx registers to a desired memory destination, according to the DMA settings.

DAU_DMA_REQEN

Reset value = 0x0000

Register Address: 0x400020A8

This register will define the slots that should contribute to the total number of ADC conversions required in the DAU_DMA_REQNUM register. MCCNUM[4:0] in the DAU_DMA_REQNUM register should be a value smaller than or equal to the total number of slots contributing to the DMA transfer, otherwise, no DMA transfer will take place.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
DMASLT_EN [15:8]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
DMASLT_EN [7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
DMASLT_EN[15:0]	15:0	Slot Enable for DMA Transfer When enabled, a completed conversion for a corresponding slot will contribute to the total number of conversions required to initiate a DMA transfer.

Revision History:

01/03/2021	Initial Revision
11/11/2021	Update Logo Added version control
24/01/2023	Corrected DAU_DMA_REQEN Register information
28/03/2023	Fixed typos, improved readability, added register bookmarks Added clarification of TRIG_SKIP and ADC_ERR bits
31/07/2024	Changed sample and hold to track and hold
19/11/2024	Corrected the calculation result in the conversion result section Corrected the description in the DAU_CONVCOMP_STATUS_INTEN table

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